2.2 Separable Differential Equations

↳ general first-order de : $\frac{dy}{dx} - f(x,y) = M(x,y) + N(x,y)\frac{dy}{dx} = 0$ (1)

independent var, dependent time

by setting $M = -f(x,y)$, $N = 1$

$$M(x) + N(y)\frac{dy}{dx} = 0 \quad (2)$$

↳ and if exclusively $x \to M$, $y \to N$, then (1) becomes (2).

said to be "separable" b/c if written in "differentiable form"

$$M(x)\,dx + N(y)\,dy = 0 \quad (3)$$

(3) can be solved by integrating M & N

→ Example: show that (a) is separable, and find equation for its integral curves.

$$\frac{dy}{dx} = \frac{x^2}{1-y^2} \quad (a) \to -x^2 + (1-y^2)\frac{dy}{dx} = 0$$

∴ separable, $M = -x^2$, $N = (1-y^2)$

$$\Rightarrow \frac{d}{dx}\left(-\frac{x^3}{3}\right) + \frac{d}{dx}\left(y - \frac{y^3}{3}\right) = 0$$

b/c chain rule, i.e. $\frac{d}{dx}\left(y - \frac{y^3}{3}\right) = (1-y^2)\frac{dy}{dx}$

∴ by integration/scale by 3 ⇒ $\boxed{-x^3 + 3y - y^3 = c}$

→ & just a plot of c values

General procedure:

given (2), let H_1 and H_2 be any antiderivatives of M, N

$\therefore H_1'(x) = M(x)$, $H_2'(y) = N(y)$

and (2) ⇒ $H_1'(x) + H_2'(y)\frac{dy}{dx} = 0$

if $y \leftarrow x$, then by chain rule:

$$H_2'(y)\frac{dy}{dx} = \frac{d}{dy}H_2(y)\frac{dy}{dx} = \frac{d}{dx}H_2(y) \quad (4)$$

$$\therefore (4) \Rightarrow \frac{d}{dx}\left(H_1(x) + H_2(y)\right) = 0 \quad (5)$$

and by integrating (5) wrt. x, we obtain

$$H_1(x) + H_2(y) = c \quad (6)$$

and any differentiable function $y = \phi(x)$ that satisfies (6) is a solution of (2)

→ & an implicit representation of the solution to (2) that satisfies (7)

$$\int_{x_0}^{x} M(s)\,ds + \int_{y_0}^{y} N(s)\,ds = 0 \quad (*)$$

w/ an initial value/condition, (7) $y(x_0) = y_0$, can obtain

2.4 Linear v.s. Nonlinear Differential Equations

* Linear DQ:
  - ↳ involve derivatives of y & y raised to first power only

* Seperable DQ:
  - ↳ can be written ~~[illegible]~~ w/ all y one side / all x to other side

Existence & Uniqueness THM of first-order DQs: → * has A solution ∧ has ONE solution

↳ if the functions p and g are continuous on an open interval $I: \alpha < t < \beta$ containing the point $t = t_0$, then $\exists\, y = \phi(t)$ s.t. —

↳ * UNIQUE

(1) $y' + p(t)y = g(t)$, for each $t \in I$, and

(2) $y(t_0) = y_0$, the satisfied initial condition

↳ arbitrary

IF (1) has a solution, it must be given by (3) $\mu(t)y = \int \mu(t)g(t)\,dt + c$

where (4) $\mu(t) = \exp \int p(t)\,dt$ * e^x

~~and reduced to~~

↳ * determines integrating factor up to a scalar dependent on lower limit of integration. Setting a lower limit at t_0, (4) becomes (5) $\mu(t) = \exp \int_{t_0}^{t} p(s)\,ds$

(6) $y = \dfrac{1}{\mu(t)}\left(\int_{t_0}^{t} \mu(s)g(s)\,ds + c\right)$

applied to (3) to obtain general solution of (1):

E & U of 1st order Nonlinear DQs:

let f, $\frac{\partial f}{\partial y}$, be continuous in $\alpha < t < \beta$, $\gamma < y < \delta$, containing point (t_0, y_0), then in some interval $\exists$ in $\alpha < t < \beta$, there exists a unique sol'n $y = \phi(t)$ of the problem

$$y' = f(t,y), \quad y(t_0) = y_0$$

2.4 co...

Example:

Use existence/uniqueness THM of first-order DEs to find an interval in which IVP

(1) $ty' + 2y = 4t^2$, (2) $y(1) = 2$

has a unique solution.

Rewriting (1): $ty' + 2y = 4t^2$

$$y' + \underbrace{\left(\frac{2}{t}\right)}_{p(t)} y = \underbrace{4t}_{g(t)}$$

$p(t)$ continuous for $t \neq 0$.

$g(t)$ continuous $\forall t$

$t > 0$ contains initial point, $\therefore$ $y = t^2 + \frac{1}{t^2}$, $t > 0$ is the unique sol'n on interval $0 < t < \infty$

Summary

$$y' + p(t)y = g(t)$$

1. assuming coefficients are continuous, $\exists$ general solution w/ constant that includes all solutions
2. equation (6) expresses this solution
3. points of discontinuity of the solution can be found w/out solving by finding discontinuities of coefficients

Example:

Use " " of first-order NONLINEAR DEs to find " "

(1) $\dfrac{dy}{dx} = \dfrac{3x^2 + 4x + 2}{2(y-1)}$, $y(0) = -1$

Observe that:

$$f(x,y) = \frac{3x^2 + 4x + 2}{2(y-1)},$$

$$\frac{\partial f}{\partial y}(x,y) = -\frac{3x^2 + 4x + 2}{2(y-1)^2}$$

continuous everywhere except on the line $y = 1$

Consequently, a continuous rectangle can be drawn about the initial point $(0,-1)$

↳ solving the equation to find solution only for $x > -2$

* Not true for nonlinear equations.

2.6 Exact Differential Equations

<u>Example:</u> Solve $2x + y^2 + 2xyy' = 0$ (1)

Not linear/separable. However, observe that $\psi(x,y)$

$\psi(x,y) = x^2 + xy^2$ has the properties

$$2x + y^2 = \frac{\partial \psi}{\partial x},$$

$$2xy = \frac{\partial \psi}{\partial y}$$

$\therefore$ (1) can be written as:

$$\frac{\partial \psi}{\partial x} + \frac{\partial \psi}{\partial y}\frac{dy}{dx} = 0 \quad (2)$$

taking y to be a function of x, we can use the chain rule to write LHS, to obtain (3) from (2)

$$\frac{d\psi}{dx}(x,y) = \frac{d}{dx}(x^2 + xy^2) = 0 \quad (3)$$

Integrating (3) to obtain

(4) $\psi(x,y) = x^2 + xy^2 = c$, the implicit solution to (1)

Let DE: $M(x,y) + N(x,y)y' = 0$

Suppose we know $\lambda(x,y)$ s.t.

$$\frac{\partial \lambda}{\partial x}(x,y) = M, \quad \frac{\partial \lambda}{\partial y}(x,y) = N$$

and $\lambda(x,y) = c$ defines $y = \phi(x)$ implicitly as a differentiable function of x. When there is a function $\lambda(x,y)$ s.t. $\lambda_x = M$ and $\lambda_y = N$, we can write

$$M(x,y) + N(x,y)y' = \frac{\partial \lambda}{\partial x} + \frac{\partial \lambda}{\partial y}\frac{dy}{dx} = \frac{d}{dx}\lambda(x, \phi(x))$$

and DE becomes

$$\frac{d}{dx}\lambda(x, \phi(x)) = 0$$

$\hookrightarrow$ it said to be <u>exact</u> eq b/c it can be expressed exactly as the derivative of a specific function. Solutions of DE are given implicitly by $\lambda(x,y) = c$

FINDING λ: $\longleftarrow$ $M_y(x,y) = N_x(x,y)$ $\longleftarrow$ IF DONE, NOT EXACT

2.6 Method of Integrating Factor

Eg. Find an int. factor for $(3xy + y^2) + (x^2 + xy)y' = 0$ & solve.

↳ Not possible because $M_y = 3x + 2y \neq N_x = 2x + y$!
(not exact)

∴ Can convert by scaling equation w/ integrating factor

Recall: $M(x,y) + N(x,y)y' = 0$

Scaling by function μ: $\mu(x,y)M(x,y) + \mu(x,y)N(x,y)y' = 0$

s.t. $(\mu M)_y = (\mu N)_x$ i.e. becomes exact.

Since M and N are given functions, μ must satisfy:

$$M\mu_y - N\mu_x + (M_y - N_x)\mu = 0$$

If μ is found, Ex. is exact. Unfortunately, is usually at least as hard to solve as original eq'n. In practice, int. factors are easy/useful in special cases. Most important situations occur when μ is a function of <u>only</u> x or y. In this case, if we assume that var to be x, then pd. μ_x reduces to $\frac{d\mu}{dx}$ and $\mu_y = 0$. Substituting,

$$\frac{d\mu}{dx} = \underbrace{\frac{M_y - N_x}{N}}\mu$$

If this is a function of x only, then $\exists$ int. factor μ that depends on x only; $\mu(x)$ can be found by solving , which is linear + separable.

3.1 Homogeneous Differential Eqns w/ Constant Coefficients

$$ay'' + by' + cy = 0 \quad (4)$$

$$\frac{d^2y}{dt^2} = f\left(t, y, \frac{dy}{dt}\right) \quad (1)$$

* second-order IVP — 2 initial conditions
  - value of function
  - slope at value

↳ linear if f has the form:

$$f\left(t, y, \frac{dy}{dt}\right) = g(t) - p(t)\frac{dy}{dt} - q(t)y \quad (2)$$

functions of independent var, not y

* ie, if f is linear in y and y'

⇓

GENERAL LINEAR: $\ddot{y} + p\dot{y} + qy = g \quad (3)$

$g = 0$? — NO → nonhomogeneous

YES → HOMOGENEOUS

solvable? — NO → no guaranteed solution to non-homogeneous variant

YES → possible to derive nonhomogeneous solution in terms of an integral.

Example

$y'' = y, \; y(0) = 2, \; y'(0) = -1$

can write as (4) where $a = 1$, $b = 0$, $c = -1$

Solutions include

$y_1 = e^t \cdot c_1$
$y_2 = e^{-t} \cdot c_2$

} and ALL linear combinations of $\langle c_1, c_2 \rangle$

Finding that which satisfies IVC:

satisfying point condition $(0, 2)$

$y = c_1 e^t + c_2 e^{-t}$ ← $2 = c_1 + c_2$

$y' = c_1 e^t - c_2 e^{-t}$

Satisfying slope condition: $t = 0$, $y' = -1$

$-1 = c_1 - c_2$

$c_1 = c_2 - 1$

$2 = c_2 - 1 + c_2$

$3 = 2c_2$

$c_2 = \frac{3}{2}, \; c_1 = \frac{1}{2}$

$$\boxed{y = \frac{1}{2}e^t + \frac{3}{2}e^{-t}}$$

SOL'N:

Looking for a function like: $y = e^{rt}$ → to be det'd.

then $y' = re^{rt}$ and $y'' = r^2 e^{rt}$. Substituting in to (4), yields

$$(ar^2 + br + c)e^{rt} = 0$$

and since $e^{rt} \neq 0 \; \forall r \; \forall t \in \mathbb{R}$,

$ar^2 + br + c = 0$ → CHARACTERISTIC EQN FOR HO.DE w/ CC.

* IIP r is a root of char. eqn. then $y = e^{rt}$ solves (4)

↳ * roots may be real + different, complex conjugates, real but repeated

$$\boxed{ay'' + by' + cy = 0} \quad (8)$$

$$ar^2 + br + c = 0 \quad (9) \longrightarrow \text{characteristic equation for } (9)$$

Assuming $r_1 \neq r_2$ are real, different roots of (9),

$y_1 = e^{r_1 t}$, $y_2 = e^{r_2 t}$. Additionally, any linear combination $\langle c_1, c_2 \rangle$:

$$\boxed{y = c_1 y_1 + c_2 y_2 = c_1 e^{r_1 t} + c_2 e^{r_2 t} \quad (10)}$$

GENERAL SOLUTION

is a solution of (8). Suppose we wish to find THE solution satisfying initial conditions:

$$y(t_0) = y_0, \quad y'(t_0) = y_0' \quad (11)$$

Substituting $t = t_0$ and $y = y_0$ in (10):

$$y_0 = c_1 e^{r_1 t_0} + c_2 e^{r_2 t_0} \quad (12)$$

Similarly, $t = t_0$ and $y' = y_0'$ in (10)'s derivative:

$$y_0' = c_1 r_1 e^{r_1 t_0} + c_2 r_2 e^{r_2 t_0} \quad (13)$$

Solving (12) + (13) simultaneously for c_1 & c_2 gives

$$c_1 = \frac{y_0' - y_0 r_2}{r_1 - r_2} e^{-r_1 t_0} \qquad c_2 = \frac{y_0 r_1 - y_0'}{r_1 - r_2} e^{-r_2 t_0}$$

3.2

Solutions ~~of~~ linear homogeneous equations, WRONSKIAN

↳ up to point: can solve some eq of form $ay'' + by' + cy = 0$

↳ Now: characterize the structure of solutions of all 2nd order linear homogeneous equations

* Differential Operator: Let p, q CFOI $\alpha < t < \beta$, α, β may equal $\pm\infty$.

Then, for any function ϕ that is twice differentiable on $\alpha < t < \beta$, differential op. L is: $L[\phi] = \phi'' + p\phi' + q\phi = 0$

LINEAR $g = 0$

* Existence & Uniqueness THM

For IVP: $y'' + py' + qy = g$

$y(t_0) = y_0, \; y'(t_0) = y_0'$

where p, q, g are CFOI that contains the point t_0. This problem has exactly ONE solution ~~$y = \phi : t \mapsto y$~~ $\phi : t \mapsto y$

* Principal of Superposition

If y_1 and y_2 are two solutions of $L[y] = 0$, (2) then linear combination (3) $c_1 y_1 + c_2 y_2$ is a solution.

states * that, given 2 solutions of eqn. (2), can construct an infinite family of solutions via (3).

Initial conditions require c_1 & c_2 s.t. ←

(4) $y_0 = c_1 y_1(t_0) + c_2 y_2(t_0)$

$y_0' = c_1 y_1'(t_0) + c_2 y_2'(t_0)$ where determinant of coefficients of system (4) $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2$ (5) behaves as:

$$W = \begin{vmatrix} y_0 & y_2 \\ y_0' & y_2' \end{vmatrix} \Big/ \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \neq 0 = c_1 = \frac{y_0 y_2' - y_0' y_2}{y_1 y_2' - y_1' y_2},$$

$$W = \begin{vmatrix} y_1 & y_0 \\ y_1' & y_0' \end{vmatrix} \Big/ \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \neq 0 = c_2 = \frac{-y_0 y_1' + y_0' y_1}{y_1 y_2' - y_1' y_2}$$

$W \neq 0 \rightarrow$ unique sol'n $\langle c_1, c_2 \rangle$ regardless of y_0 & y_0'

and solution given by

if $W = 0$, many conditions unsatisfied regardless of $\langle c_1, c_2 \rangle$

3.5

Non-homogenous equations; method of undetermined coefficients

↳ Considering NON-HOMOGENOUS SECOND-ORDER ~~LINEAR~~ DQ:

$$L[y] = y'' + py' + qy = g \quad (1)$$

where p, q, and g are CFOI I. $L[y] = 0$, the corresponding homogenous dq to (1) may be used to characterize the general solution/structure thereof:

↳ * If y_1 and y_2 are 2 solutions of NHLDQ (1), then $y_1 - y_2$ is a solution of corresponding homogenous dq (2). If, additionally, y_1 and y_2 form a fundamental set of solutions of (2), then $y_1 - y_2 = c_1 y_1 + c_2 y_2$ (3)

↳ * The general solution of NHLDQ (1) can be written in the form:

$$\phi : t \to y = c_1 y_1 + c_2 y_2 + Y$$

* any sol'n to (1)

↳ * form the fundamental set of solutions of the corresponding HLDQ (2)

Method of undetermined coefficients.

requires an initial assumption about the form of a particular sol'n Y, w/ unspecified coefficients.

↳ then substitute the assumed expression into NHLDQ (1), and attempt to determine coefficients that satisfy the equation.

WORKED? $\xrightarrow{\text{YES}}$ Y

↓ NO

No solution exists for the assumed form.

* typically only used for constant HLDQ.